

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 33

Title: Blood Disorders:
Hemoglobinopathies, Thalassemia and Hemophilia

Dr. Mohamed Idries

Email: midries@sgu.edu

Acknowledgement: Dr. Sharmila Upadhya

St Georges University ©

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Lecture: Hemoglobinopathies

SOM.MK.III.BPM1.1.CPR.3.GNET.0180	Review molecular mechanism of sickle cell anemia from the FTM module.
SOM.MK.III.BPM1.1.CPR.3.GNET.0181	Interpret data and differentiate between RFLP and ASO blot in the interpretation of inherited hemoglobin disorders.
SOM.MK.III.BPM1.1.CPR.3.GNET.0182	Explain the effect on the protein due to a single base substitution in HbS and HbC.
SOM.MK.III.BPM1.1.CPR.3.GNET.0183	Analyze the difference between sickle cell anemia (homozygous) and sickle cell trait (heterozygous/carrier). Indicate the role of heterozygote advantage of the sickle cell trait in parts of the world where malaria is endemic
SOM.MK.III.BPM1.1.CPR.3.GNET.0184	Explain the molecular basis for the clinical features of HbS and HbC and the use of hydroxyurea in management of HbS.
SOM.MK.III.BPM1.1.CPR.3.GNET.0185	Discuss the HbC mutation and the phenotype of the HbC carrier, HbC homozygotes and HbS/HbC individuals.
SOM.MK.III.BPM1.1.CPR.3.GNET.0186	Compare the genomic arrangement of the α -globin and β -globin genes
SOM.MK.III.BPM1.1.CPR.3.GNET.0187	Compare, and contrast the most common mutations in α -thalassemia and β -thalassemia, explaining the thalassemia trait (heterozygous state).
SOM.MK.III.BPM1.1.CPR.3.GNET.0188	Explain the concept of allelic heterogeneity and compound heterozygote in relation to β -thalassemia.
SOM.MK.III.BPM1.1.CPR.3.GNET.0189	Analyze the genotype-phenotype correlations in α -thalassemia and β -thalassemia.
SOM.MK.III.BPM1.1.CPR.3.GNET.0190	Explain the mode of inheritance of hemophilia A and B, the common clinical features and the most common mutation (Intron inversion) in hemophilia A.
SOM.MK.III.BPM1.1.CPR.3.GNET.0191	Explain the genetic basis of manifestations in a hemophilia carrier female (skewed X inactivation)
SOM.MK.III.BPM1.1.CPR.3.GNET.0192	Calculate the recurrence risk in families with hemophilia A (When mother is a carrier or when father is affected), HbS, HbC, alpha-thalassemia and beta-thalassemia

Recommended Reading

- Korf & Irons: Clinical Snapshot 7.1 – Sickle Cell Anemia – p131
- Korf & Irons: page 205 – discussion of thalassemia
- Self-Assessment Review Questions: Korf & Irons: Chapter 13 questions
- Practice question 3.1 - 3.7, 31.1 in Lippincott's Biochemistry 8th edition
- Lippincott's Biochemistry 9th edition: Chapter 3: Globin gene organization and Hemoglobinopathies;
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lww.healthlibrary.com/book.aspx?bookid=3402>
- FTM - Molecular mechanisms in inherited disorders – HbS
- Review FTM – Gene regulation
- Review FTM – Hemophilia (X-linked) transmission and risk assessment; Autosomal recessive disorders risk assessment

Blood Disorders: Hemoglobinopathies, Thalassemia and Hemophilia

Dr. Mohamed Idries

midries@sgu.edu

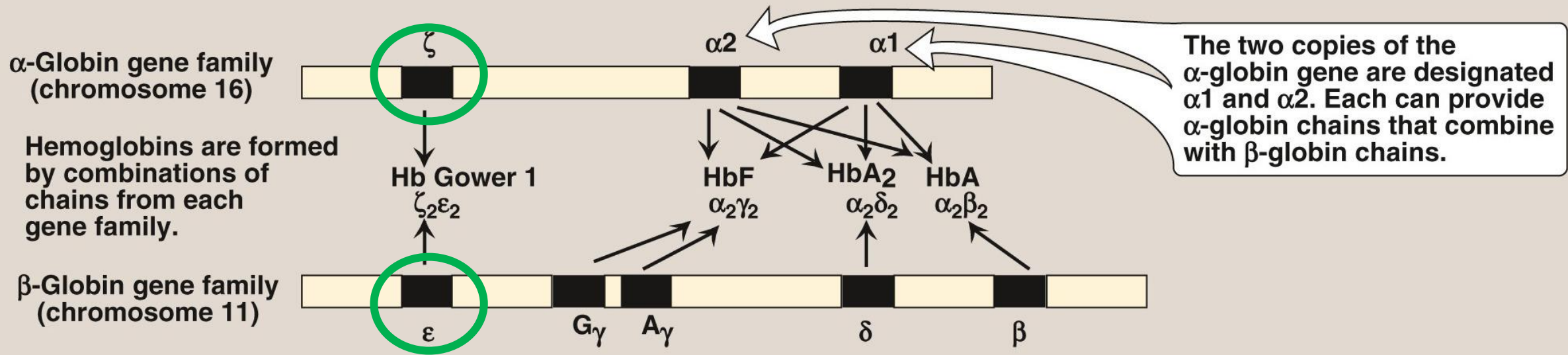
Department of Biochemistry and Genetics

<https://emedicine.medscape.com/article/955496-overview>

<https://medlineplus.gov/genetics/condition/beta-thalassemia/#inheritance>

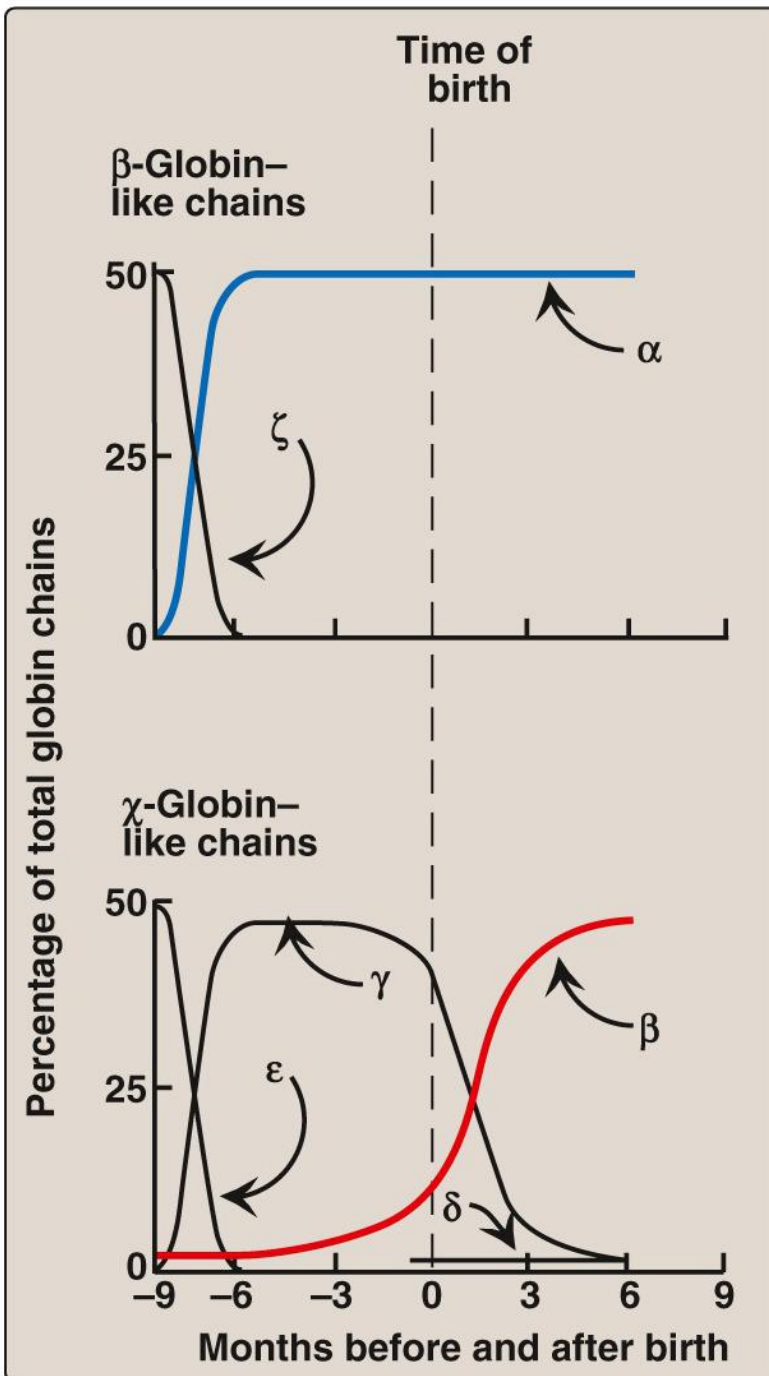
<https://rarediseases.info.nih.gov/diseases/871/beta-thalassemia>

Globin gene organization



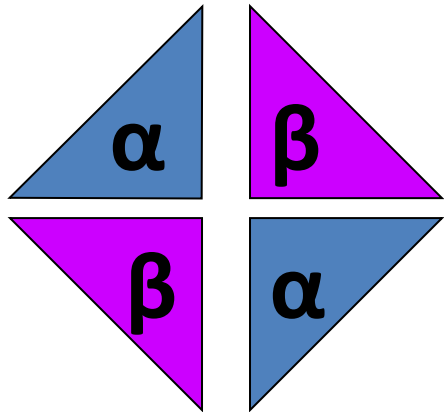
- **α-like globin genes on chromosome 16; β-like globin genes on chromosome 11**
- Various hemoglobin species formed throughout development from these genes
- 2 types of α-globin genes (*HBA1* and *HBA2* genes) (**4 α-globin genes in normal individual**)
- 1 β-globin gene (*HBB*; **2 β-globin genes in normal individual**)
- Zeta and epsilon (**embryonic globin chains**)

Globin gene expression

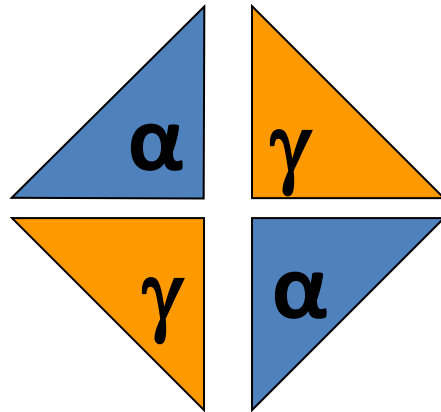


Timeline of expression of globin genes from early fetal development to birth and in the first year of life

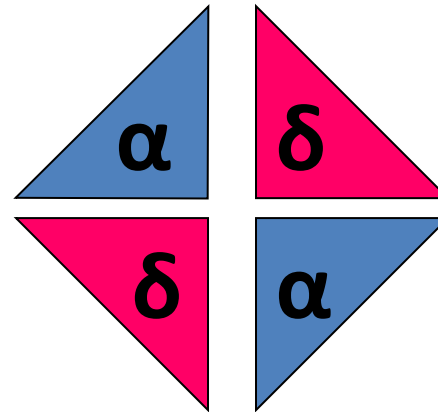
Composition of hemoglobin



HbA
90-95%



HbF
1-2%



HbA₂
2-3%

Form	Chain composition	Fraction of total hemoglobin
HbA	$\alpha_2\beta_2$	90%
HbA ₂	$\alpha_2\delta_2$	2%–3%
HbF	$\alpha_2\gamma_2$	<2%
HbA _{1c}	$\alpha_2\beta_2$ -glucose	4%–6%

- Hb consist of two different pairs of globin chains (tetramers) with heme
- Embryonic hemoglobins have ζ (zeta) chains and ϵ (epsilon) chains
- Fetal hemoglobin (HbF): 2 α chains and 2 γ chains ($\alpha_2\gamma_2$)
- In adults, hemoglobin A ($\alpha_2\beta_2$) and hemoglobin A₂ ($\alpha_2\delta_2$)

Hemoglobinopathies and Thalassemia

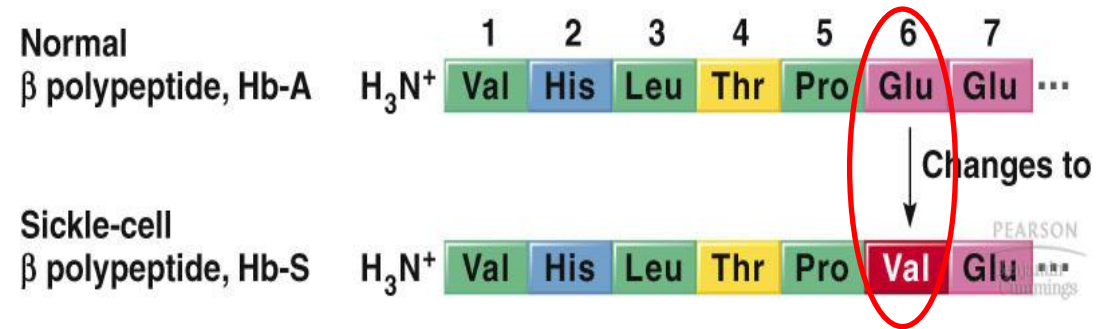
Hemoglobinopathies: Genetic disorders caused by production of structurally abnormal hemoglobin or by synthesis of insufficient amounts of normal hemoglobin

Hundreds of mutations known, but we shall look at **four** of these in detail

Hemoglobinopathies → **Qualitative** change:
Mutation in nucleotide sequence of globin chain

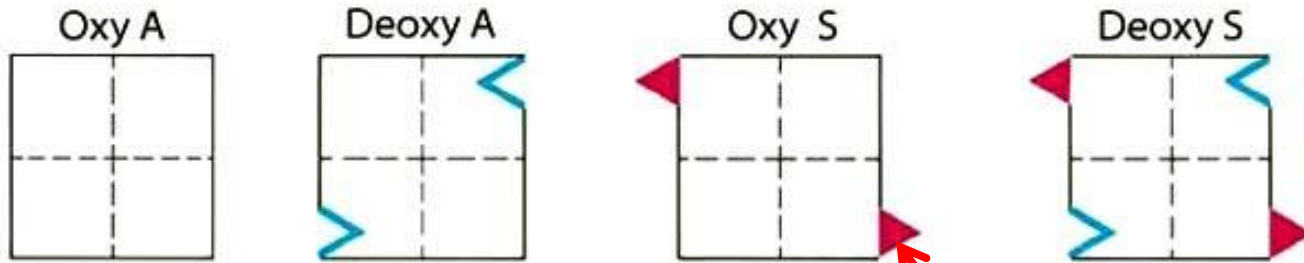
Thalassemia → **Quantitative** change:
Decreased or absent globin chain synthesis

Hemoglobin S



- Point mutation in β-globin gene (*HBB*) on 6th codon
- Glutamic Acid substituted by **Valine** at position 6 of β-globin chain
- HbS moves slower to anode than HbA in electrophoresis
- Homozygous inheritance (HbSS) causes sickle cell disease
 - NO production of HbA
- **Autosomal recessive disorder**
- Painful episodes 'sickle cell crisis' and hemolytic anemia
- Reduced life-span of RBCs
- Are there manifestations of sickle cell anemia during fetal life???

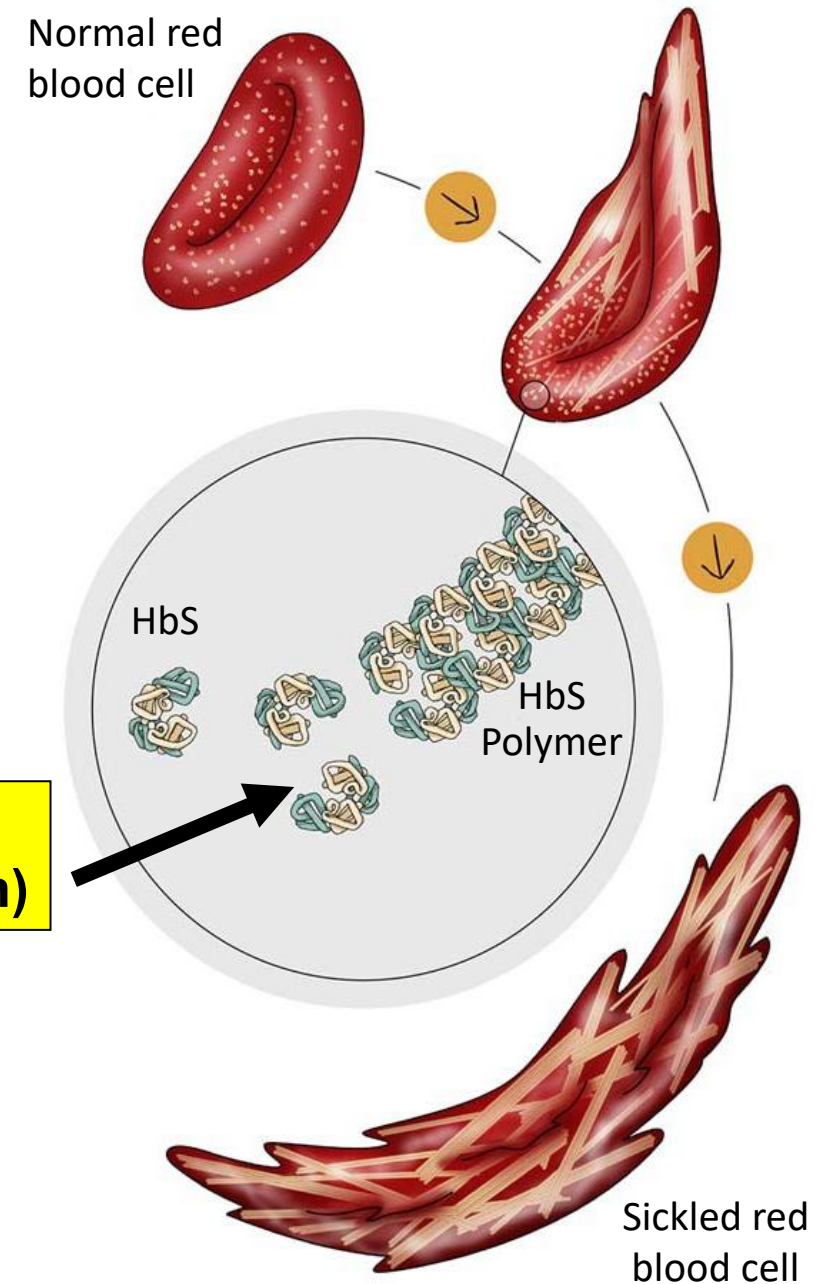
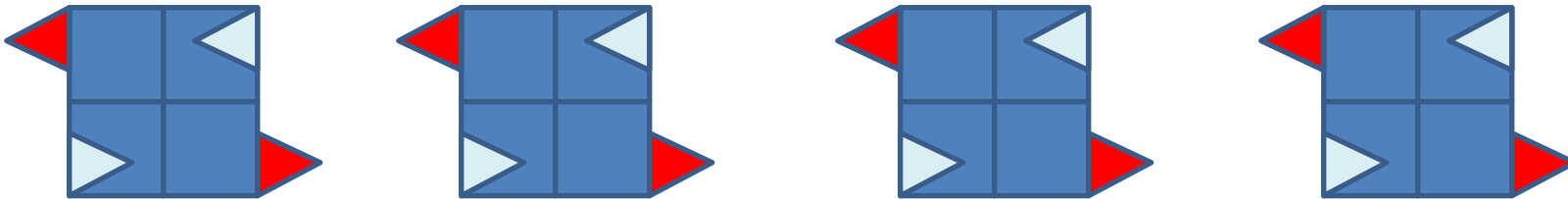
Changes in deoxy HbS



Hydrophobic 'sticky' patch in HbS (Glu→Val)

Poor solubility in deoxy state

Novel Property (Sickling/polymerization)



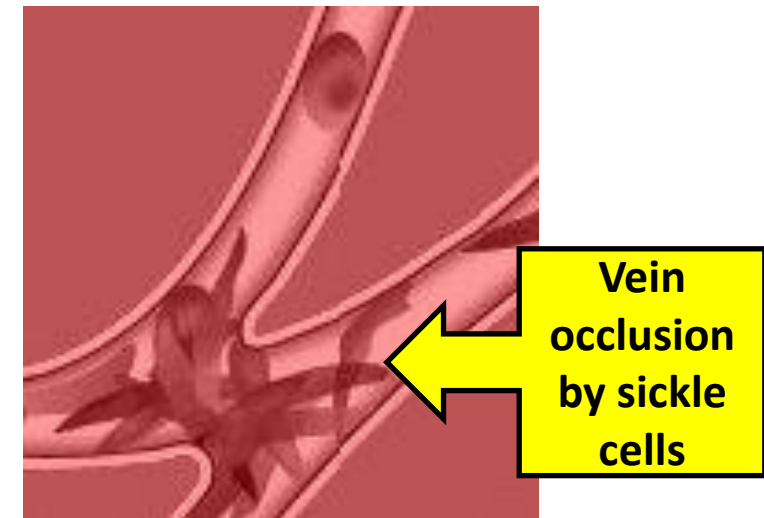
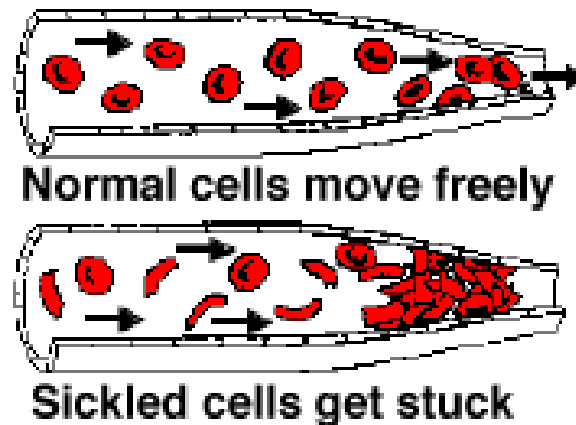
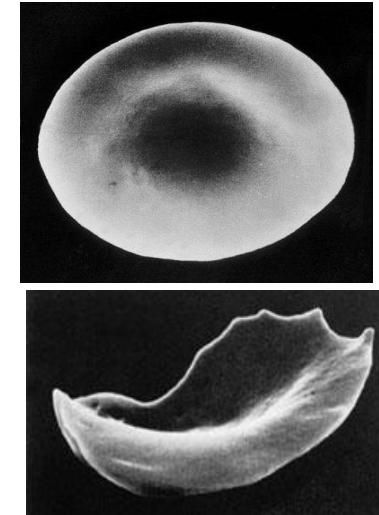
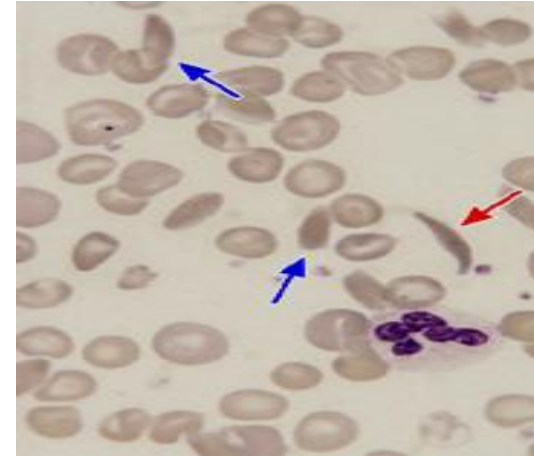
Clinical features and Appearance of RBCs in sickle cell anemia

Causes of sickling (factors that increase deoxyHb):

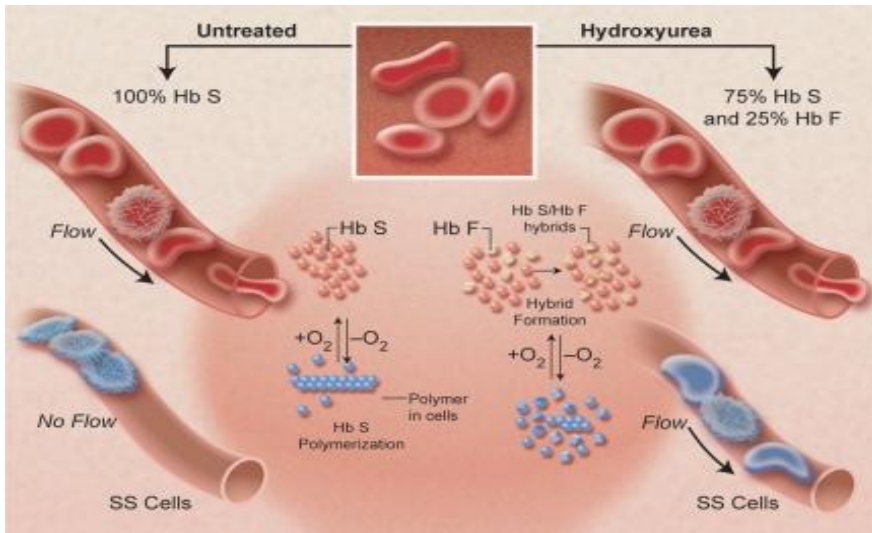
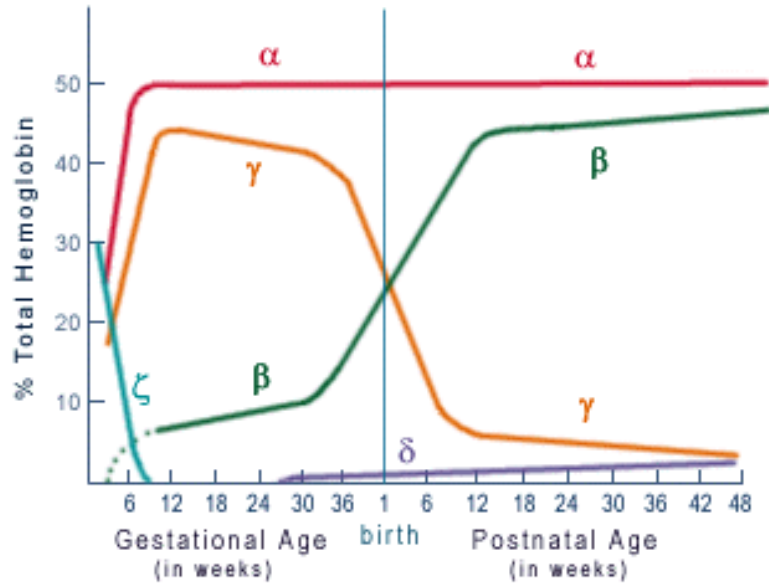
1. **Acidosis**
2. **Volume depletion**
3. **Hypoxemia**

Painful crises caused by anemia, hemolysis and vaso-occlusive ischemia in abdomen and long bones:

1. **Dactylitis**
2. **Acute chest syndrome**
3. **Avascular necrosis of femoral head**
4. **Autosplenectomy**
5. **Stroke**



HbS Treatment: Fetal hemoglobin inducers



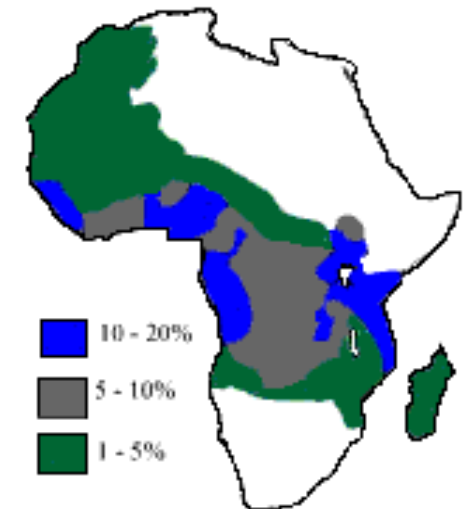
- **Methylation** of CG promoters repress fetal globin (γ) transcription after birth
- Drugs increase fetal globin (γ) expression:
 - 5-azacytidine (Decitabine; demethylating agent)
 - **Hydroxyurea**, Butyrate compounds (**inhibit** histone deacetylation)
- Alter **epigenetic gene** regulation and change histone acetylation → Switching **on γ -globin synthesis** → **‘Derepression’** of γ -globin gene → **More HbF** formation
- **Hydroxyurea: Increases Hb F levels**
 - HbF has a higher affinity for O₂ → inhibits sickling
- Mixed hybrid $\alpha_2\beta^S\gamma$ less likely to polymerize than hemoglobin S

Emerging Therapies in Sickle Cell Disease

- **Casgevy** (exagamglogene autotemcel):
 - CRISPR/Cas9 gene-editing technology
 - Modifies patient's hematopoietic stem cells
 - Prevents the production of the BCL11A protein, which normally inhibits HbF production after birth
 - Casgevy allows more HbF to be generated
- **Lyfgenia** (lovotibeglogene autotemcel):
 - Uses a lentiviral vector to introduce a modified form of the hemoglobin gene into the patient's hematopoietic stem cells
 - This variant of hemoglobin functions similarly to normal adult hemoglobin (HbA)

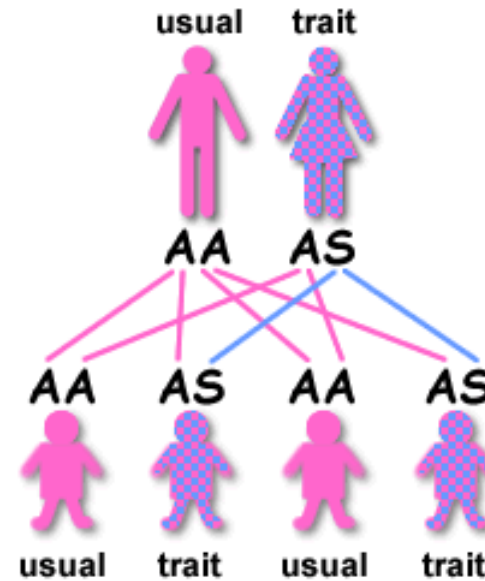
Sickle cell trait- Heterozygous carrier HbAS

- Heterozygous (HbAS) sickle cell trait (1 normal β -globin gene and 1 β^s -globin gene)
- Produce both normal and abnormal hemoglobin
- Generally **benign**
- Develop sickling crisis in **low oxygen saturation**
 - Deep sea diving and unpressurized aircraft
 - Extreme exercise such as army training
- **Africa, Mediterranean basin, Middle East and India**
 - High carrier frequency
- Heterozygotes - **survival advantage (Heterozygote advantage)** against **falciparum malaria**

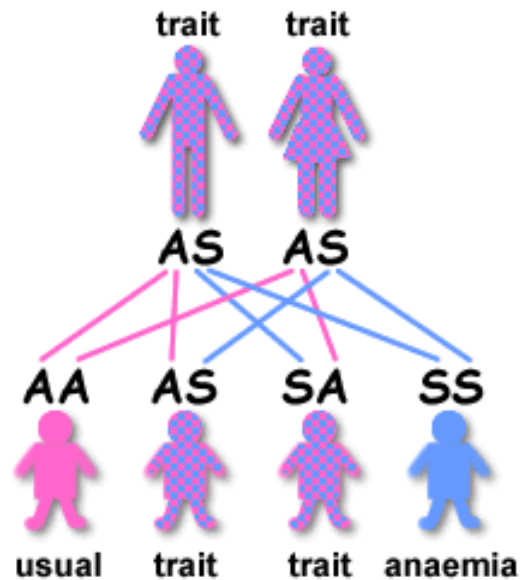


Inheritance

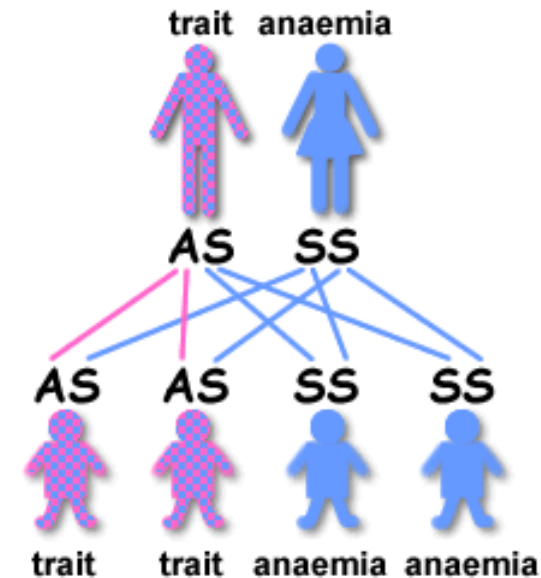
- **Autosomal recessive**
- 2 copies of gene for Hb
- HbS - Recessive
 - S - Sickle
 - A - Normal



Review from FTM module



$\frac{1}{4}$ (25%) risk to next child



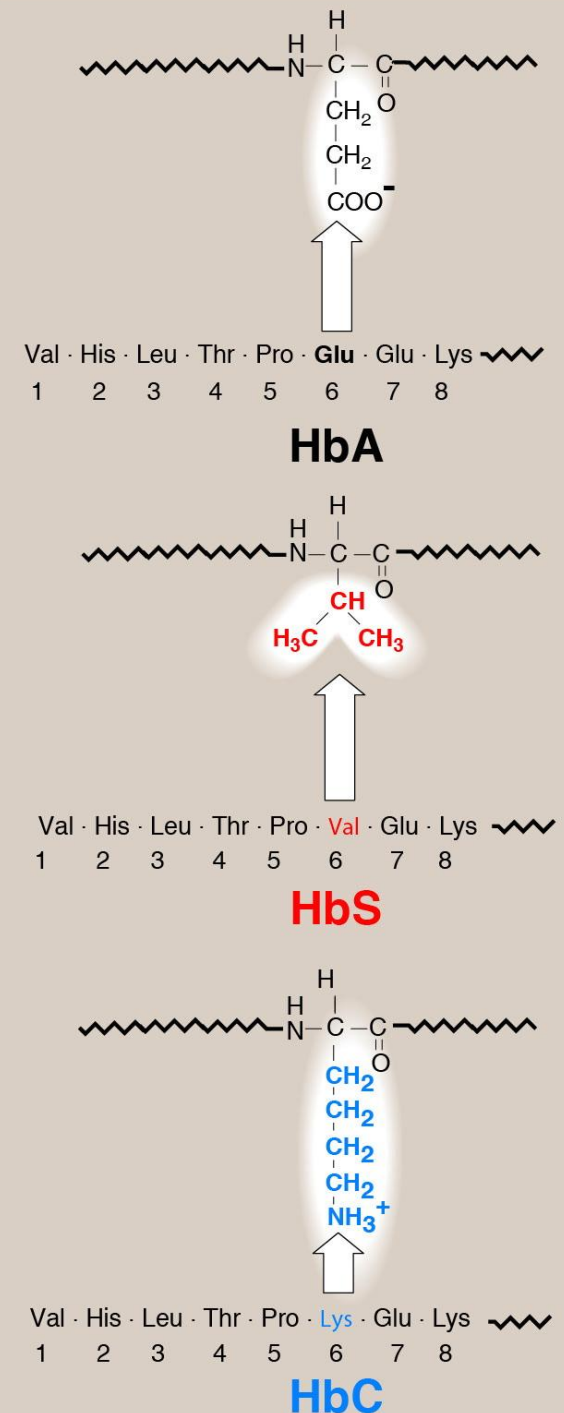
Pseudo-dominant inheritance
(high carrier frequency)

Hemoglobin C

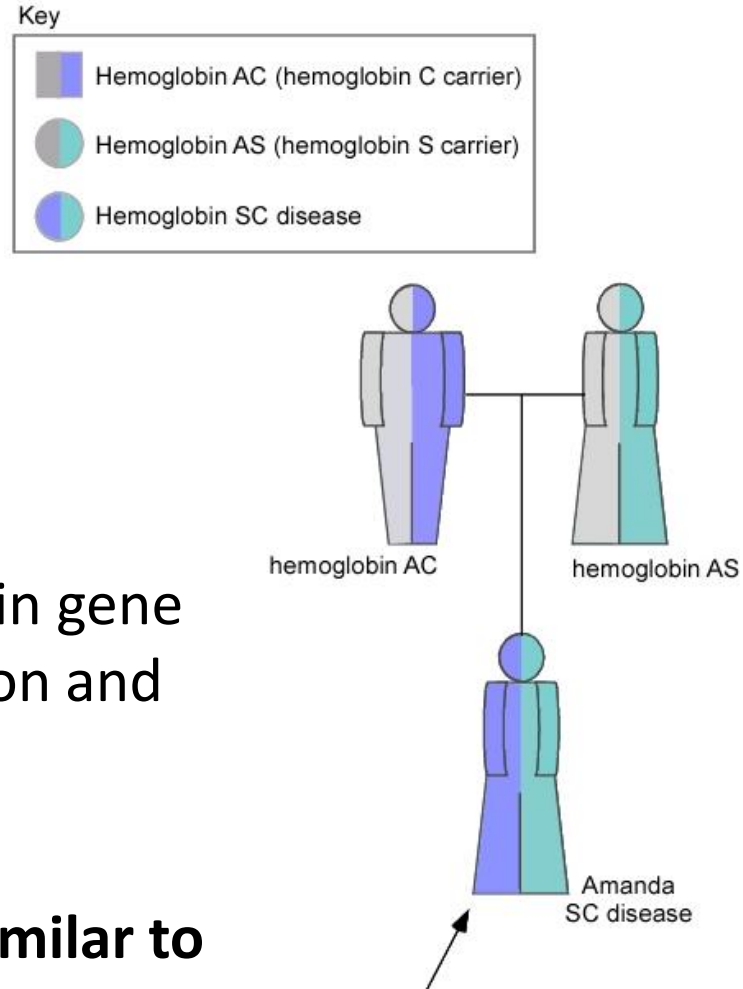
b) β -chain

	Amino acid position									
	1	2	3	6	7	26	63	67	121	146
Normal	Val	His	Leu	Glu	Glu	Glu	His	Val	Glu	His
Hb variants:										
Hb-S	Val	His	Leu	Val	Glu	Glu	His	Val	Glu	His
Hb-C	Val	His	Leu	Lys	Glu	Glu	His	Val	Glu	His

- **Point mutation 6th codon** of β -globin gene (HBB gene) $\rightarrow$ **Missense mutation** (Glu $\rightarrow$ **Lys**)
- HbC moves slowest to anode (positively charged electrode) on hemoglobin electrophoresis
- HbC homozygotes have **mild hemolysis**
- HbC - Lower solubility than HbA
- HbA1C – Compare HbC and HbA1C



HbSC disease



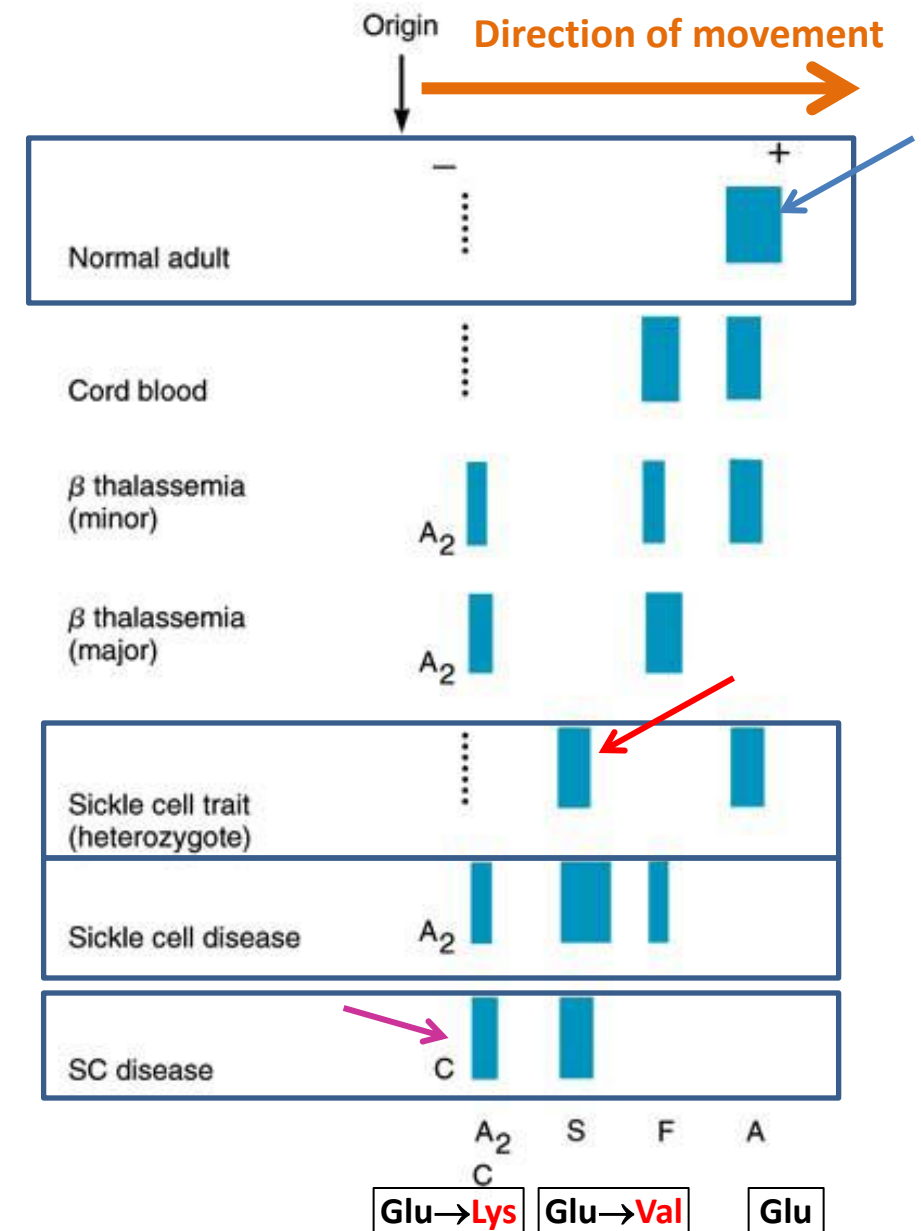
Two different mutations of β -globin gene
One β -globin gene has 'S' mutation and
other has 'C' mutation

**HbSC: Sickling episodes similar to
sickle cell disease (milder)**

**What is the risk of HbSC
to the next child?**

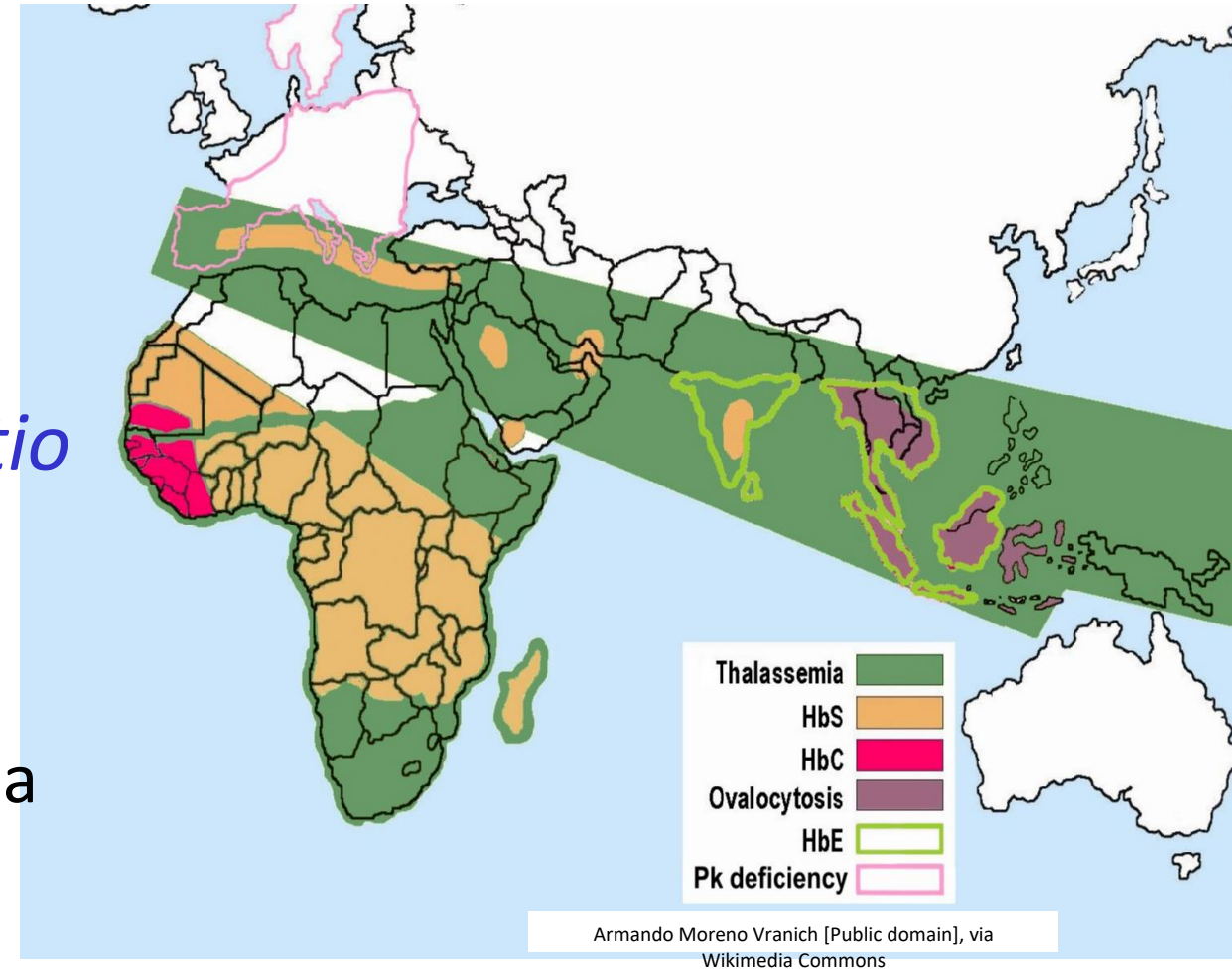
Hemoglobin electrophoresis: HbS, HbAS and HbSC

DNA diagnostic tests:
ASO and RFLP

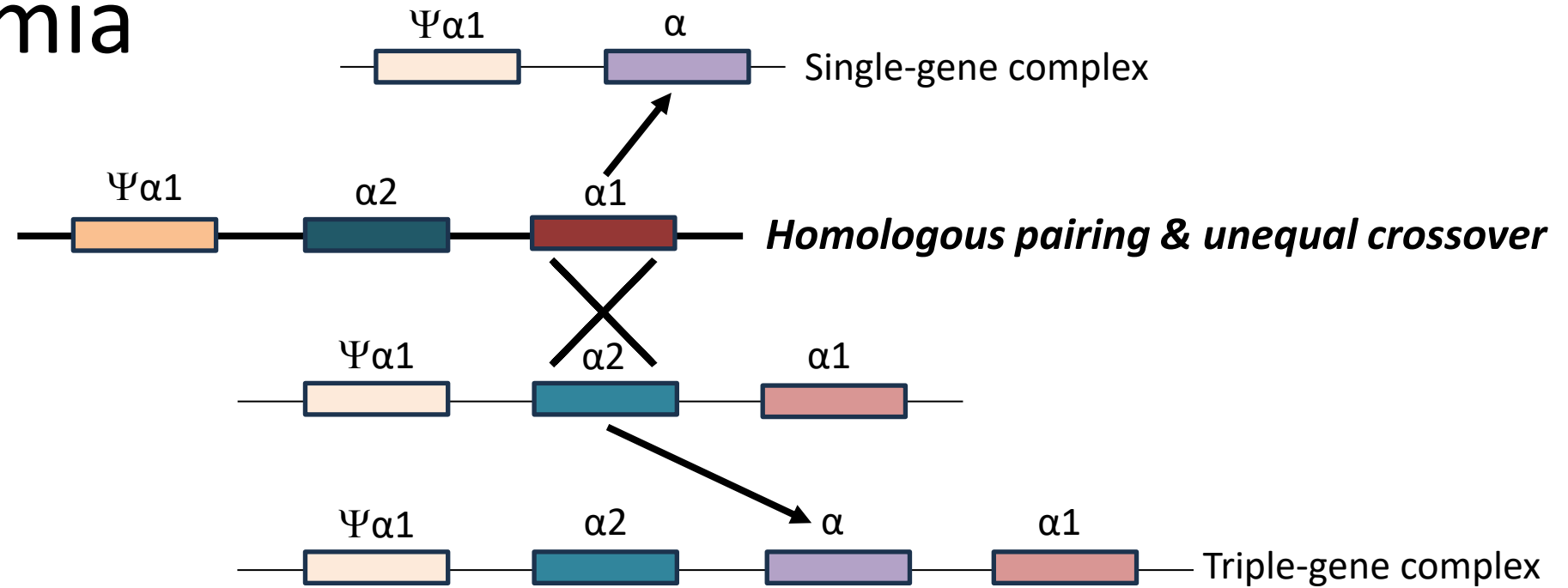


Thalassemias

- Reduced globin chain synthesis
 - Alpha thalassemia - reduced alpha chain synthesis
 - Beta thalassemia - reduced beta chain synthesis
- *Imbalance in Alpha: Beta chain ratio*
 - Normal ratio is 1:1
- *Thalassemia* (Greek): *thalassa* (sea) and *haima* (blood)
- South-east Asia, Africa, Mediterranean
- Heterozygotes protected against malaria



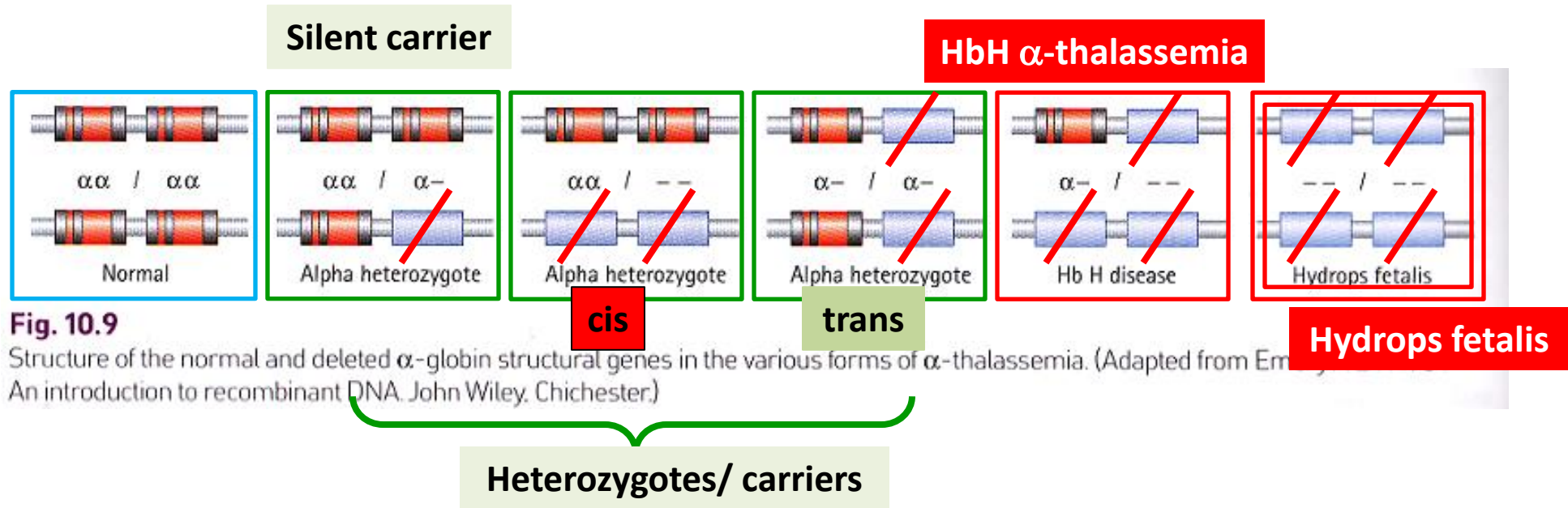
α -Thalassemia



Adapted from Orkin 1987

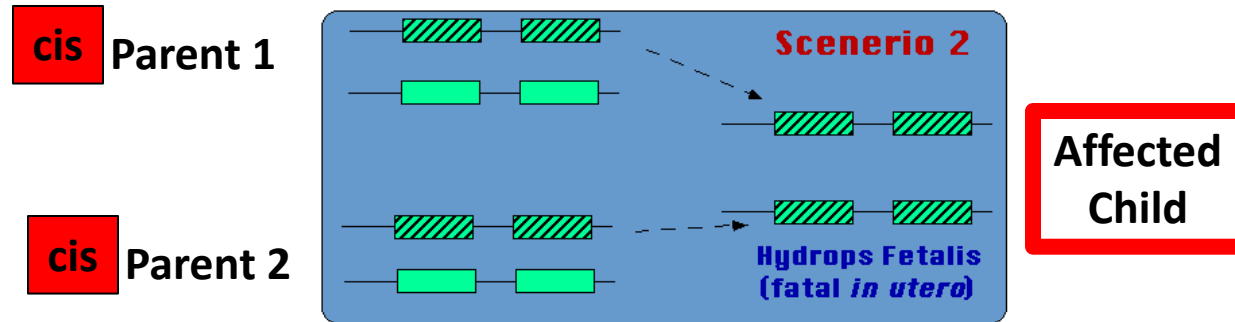
- Reduced α -globin chain synthesis
 - Relative excess of beta-globin or gamma-globin chains
- **Deletion of entire α -globin gene**
 - Unequal cross-over during homologous recombination during meiosis
- 2 α -globin genes: **4 copies** of α -globin genes (*HBA1* and *HBA2*)
 - Complex inheritance

Genetic basis of α -thalassemia



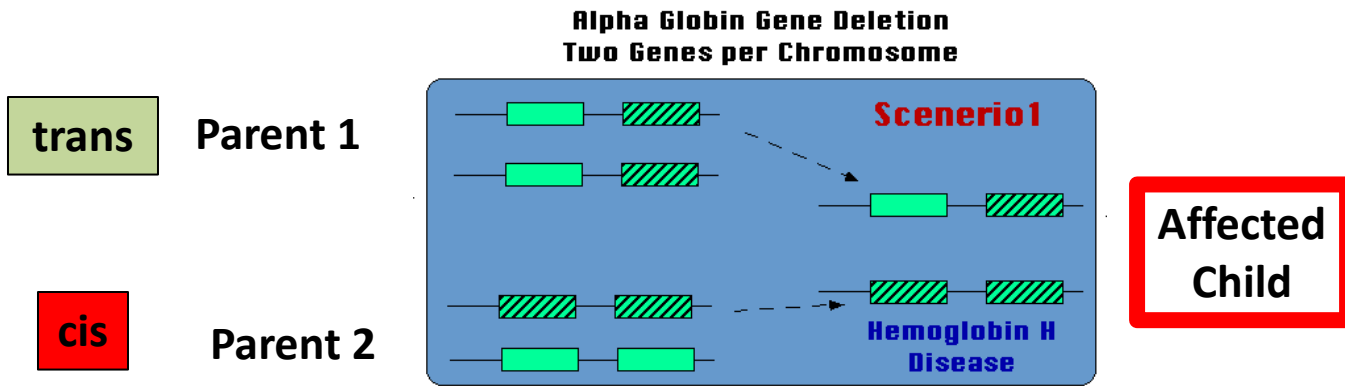
- Two forms of α -thalassemia:
 - Hydrops fetalis (Hb Bart)
 - Hemoglobin H disease
- Carrier: Deletion of **two α -globin genes**
 - $\alpha\alpha/--$ in **cis** configuration or
 - $\alpha-/-\alpha$ in **trans** configuration

Hemoglobin Bart hydrops fetalis (Hb Bart) syndrome

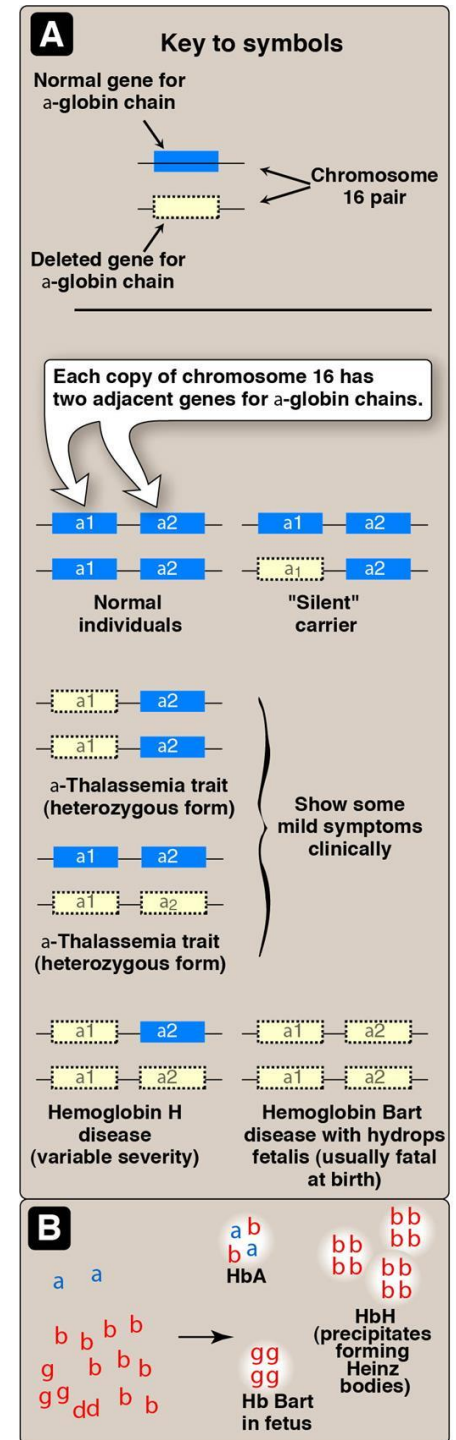


- Most severe form of α -thalassemia
- **Hb Bart**: Aggregation of γ_4 tetramers (Due to absence of α -globin chains)
 - Can be detected prenatally
- Hb Bart extreme affinity for oxygen ($10\times > \text{HbA}$) $\rightarrow$ O₂ not released to peripheral tissue $\rightarrow$ severe hypoxemia $\rightarrow$ high output cardiac failure
- Fetal onset of generalized edema, ascites, pleural and pericardial effusions, and severe anemia
- Stillbirth or death in neonatal period
- All **four α -globin alleles deleted** (inactivated)
- **No detectable HbF or HbA**

Hemoglobin H (HbH) disease



- Presents in infancy or childhood
- Mild-to-moderate microcytic hypochromic hemolytic anemia (chronic hemolysis)
- Hepatosplenomegaly (increased splenic sequestration)
- Mild thalassemia-like bone changes may be present
- **HbH**: Aggregation of β_4 tetramers (deficiency of α -globin chains)
- May also have some HbBart (aggregation of γ_4 tetramers)
- Deletion of **three of four α -globin alleles**
- **HbA levels may be low; or low normal**

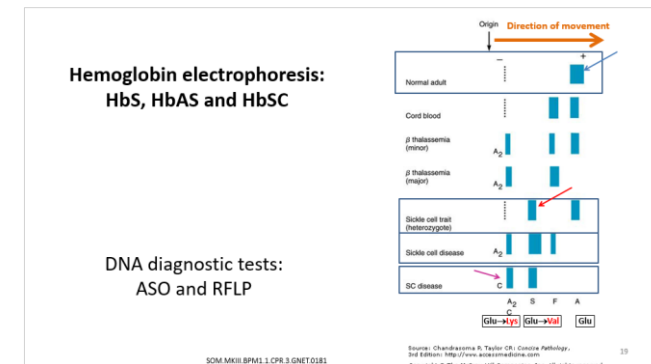


β -thalassemia

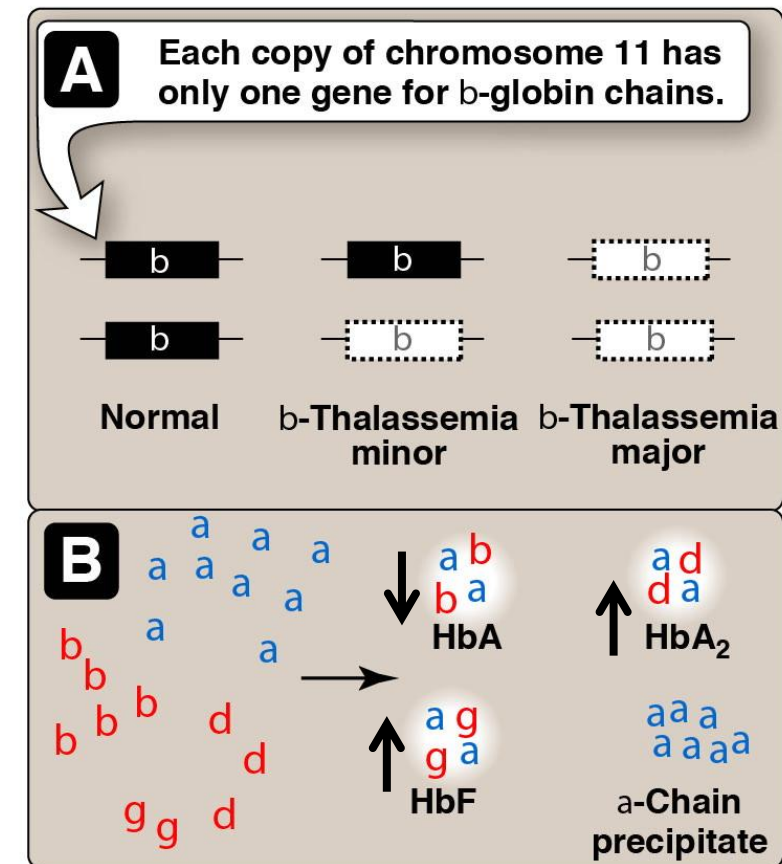
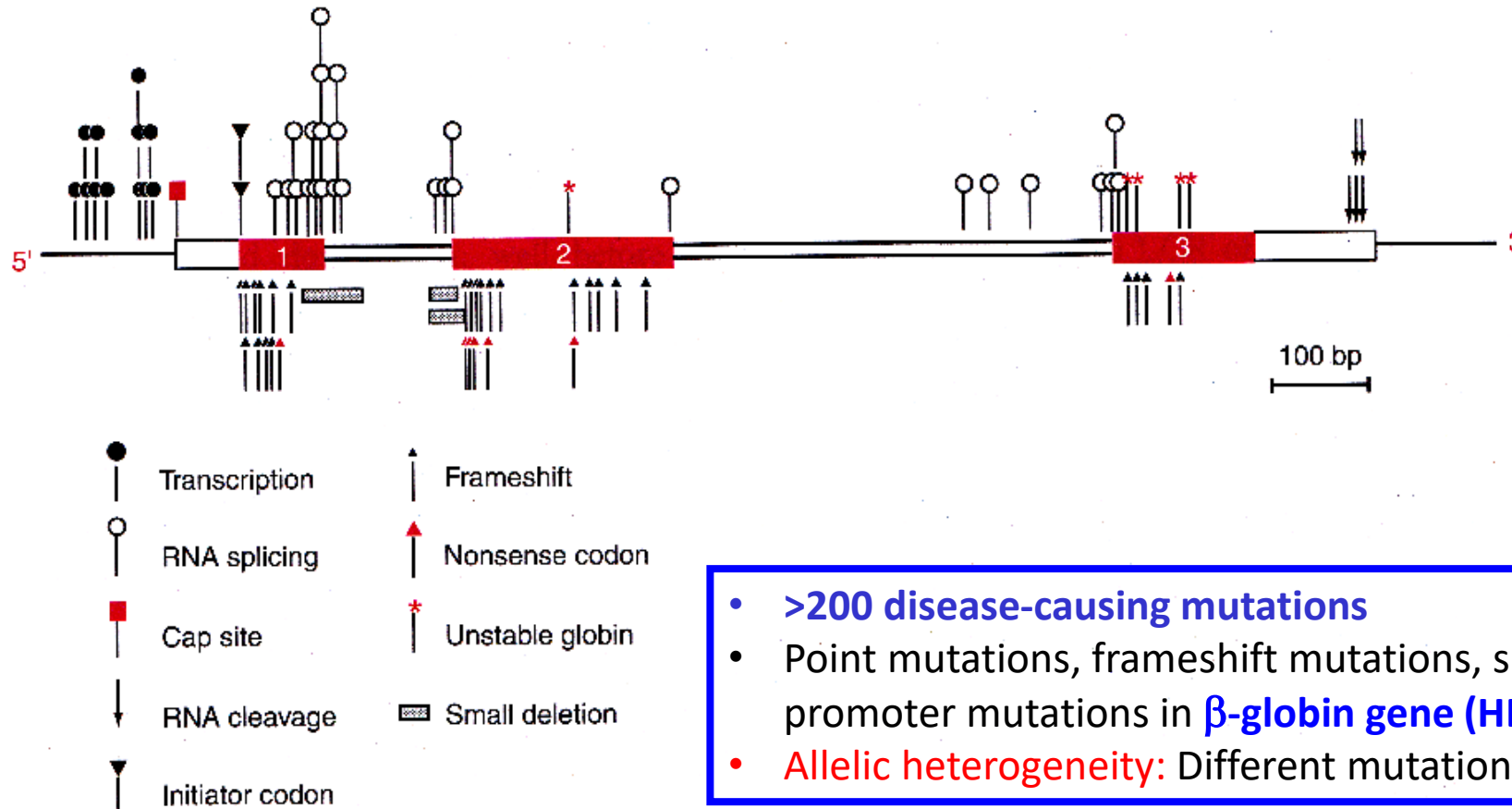
- **Autosomal recessive** inheritance
- Different mutations of beta-globin gene (**allelic heterogeneity**)
 - **Compound heterozygotes** (*HBB* gene)
- Variable severity of disorder
- Types of beta globin mutation
 - **β^+ mutation**: reduced gene expression
 - **β^0 mutation**: complete suppression of gene expression
- Absent or reduced synthesis of **β -globin chains** of Hb
- Excessive α -globin chains precipitate and **severe hemolytic anemia**
 - **Excess α -globin chains** bind to RBC membranes causing membrane damage
 - Form toxic aggregates resulting in hemolysis
 - (Remember: they do not form tetramers)

β -thalassemia: Three forms

- **β -thalassemia major** ('Cooley's Anemia' or 'Mediterranean Anemia')
 - Homozygotes or compound heterozygotes for β^0 or β^+ genes
 - **Two severe mutations**
 - **Very low or absent HbA levels** – Absent or very little β globin synthesis
 - High HbA2 and HbF levels (Compensatory)
- **β -thalassemia intermedia**
 - Homozygotes or compound heterozygotes (different β^+ mutations)
 - **One severe mutation, second mutation less severe or two less severe mutations**
 - **Low HbA levels** – Some β globin synthesis
 - High HbA2 and HbF levels (Compensatory)
- **β -thalassemia minor** (Carrier or trait or Heterozygotes)
 - **Heterozygotes** (one normal and one mutant β -globin gene)
 - Almost normal HbA levels



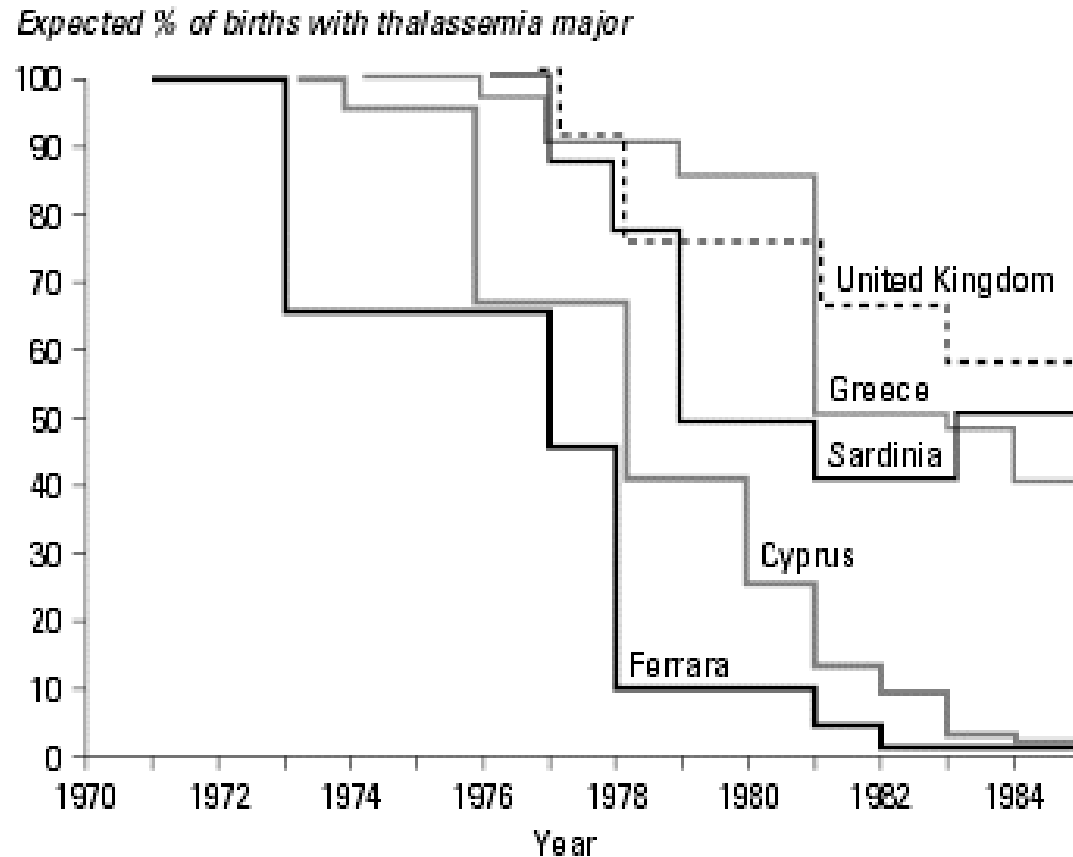
Allelic heterogeneity in β -thalassemia



- **>200 disease-causing mutations**
- Point mutations, frameshift mutations, splice site mutations, promoter mutations in **β -globin gene (HBB gene)**
- **Allelic heterogeneity:** Different mutations in same gene

Figure 11-13. Representative point mutations that cause β -thalassemia. Note the distribution of mutations throughout the gene and that the mutations affect virtually every process required for the production of normal β globin. More than 100 different β -globin point mutations are associated with simple β -thalassemia. (Redrawn from Kazazian HH [1990] The thalassemia syndromes: Molecular basis and prenatal diagnosis in 1990. In Miescher LA, Jaffé ER [eds] Seminars in Hematology 27. WB Saunders, Philadelphia, pp. 209-228.)

Effect of prenatal diagnosis and carrier screening



Source: Modified from Modell and Bulzhenkov (1998, 244).

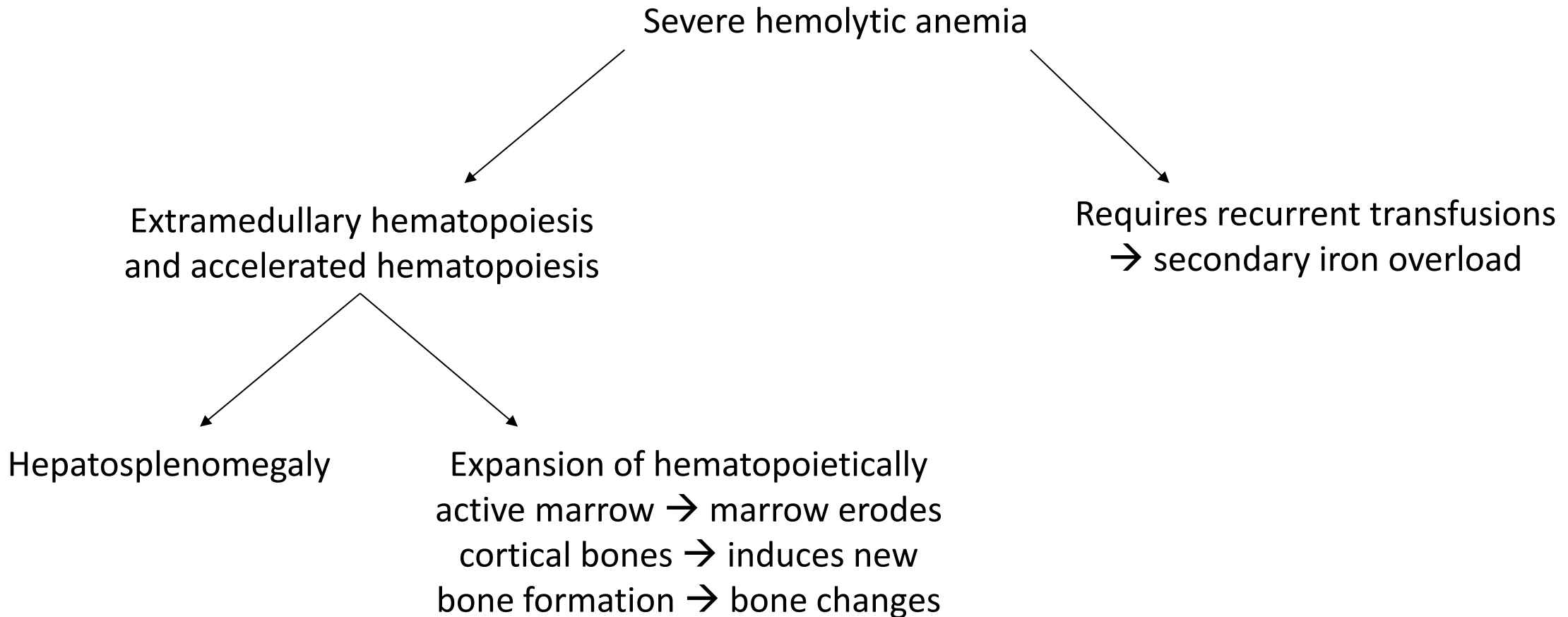
- **Carrier screening** in Mediterranean populations
- Ethical concerns about termination of pregnancy and forced participation have been balanced against severity of condition
- Screening successful in reducing incidence of disorder

High carrier frequency in Mediterranean, North Africa and Middle East

Decline in thalassemia in various populations after initiation of prenatal diagnosis in 1972

β -thalassemia major phenotype

β^0 homozygotes (thalassemia major) present at about **six months – 2years** as expression of γ -globin genes decreases

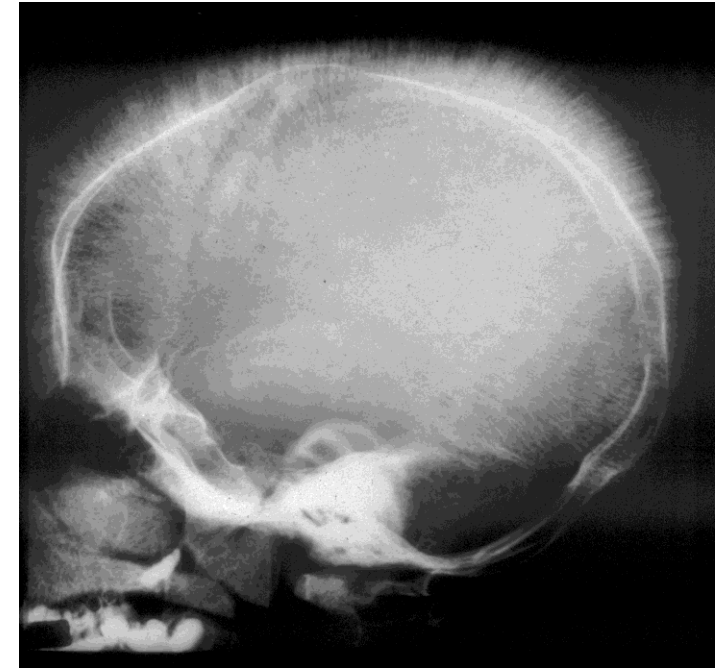


Bone changes in thalassemia

(More common in beta-thalassemia; may be seen in alpha-thalassemia)



Frontal bossing, malar prominence: Indicate
Extramedullary erythropoiesis



"Hair-on-end" skull radiograph

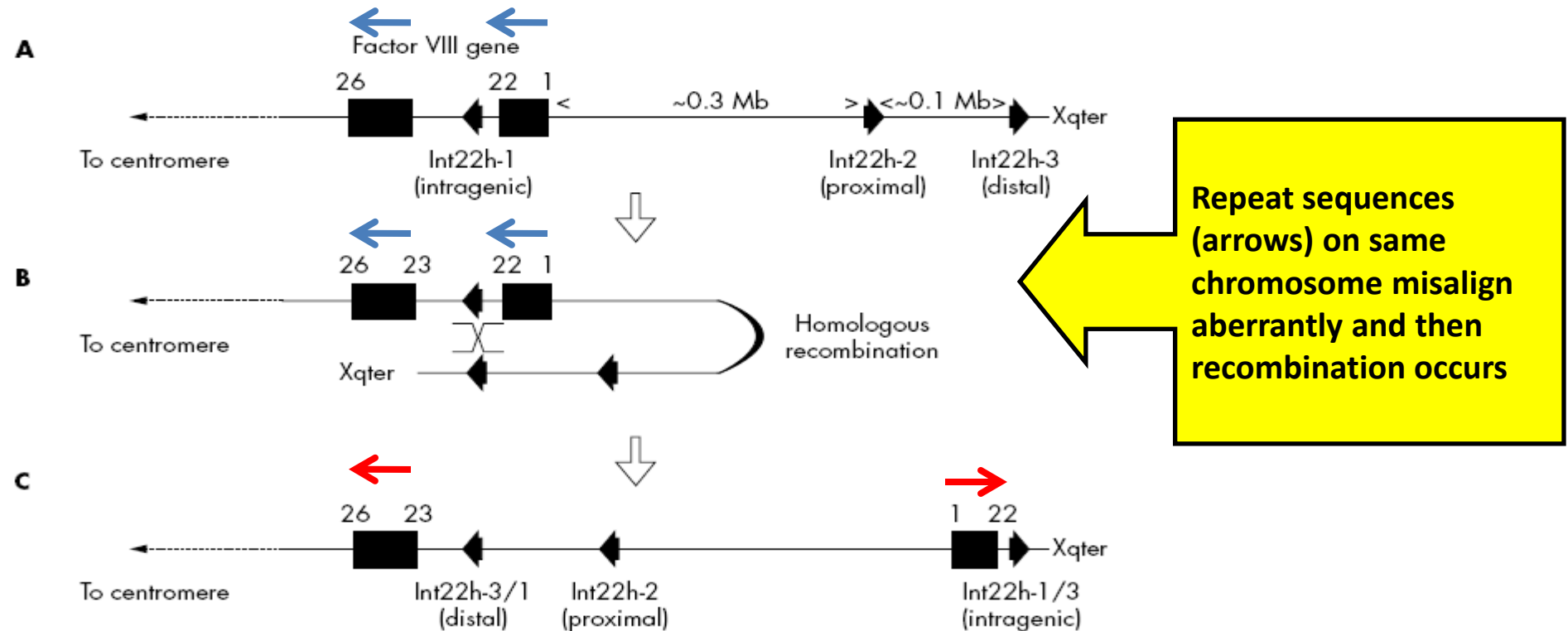
Bone marrow tries to compensate and expands to perform erythropoiesis and leads to bone deformity and fractures; Erythropoiesis in flat bones is stimulated (**extramedullary erythropoiesis**)

Hemophilia A and B

- **X-linked recessive** bleeding disorders
 - Deficiency of Factor VIII (hemophilia A) or
 - Factor IX (hemophilia B)
- Hemorrhage into joints and muscles, easy bruising, and prolonged bleeding from wounds
- Hemophilia A and B clinically similar
 - Distinguished by assays of factor VIII and factor IX
 - **Prolonged APTT**; **PT: normal**; **Bleeding time: normal**
- Incidence:
 - 1 in 5,000-10,000 male births for hemophilia A
 - 1 in 50,000 male births for hemophilia B



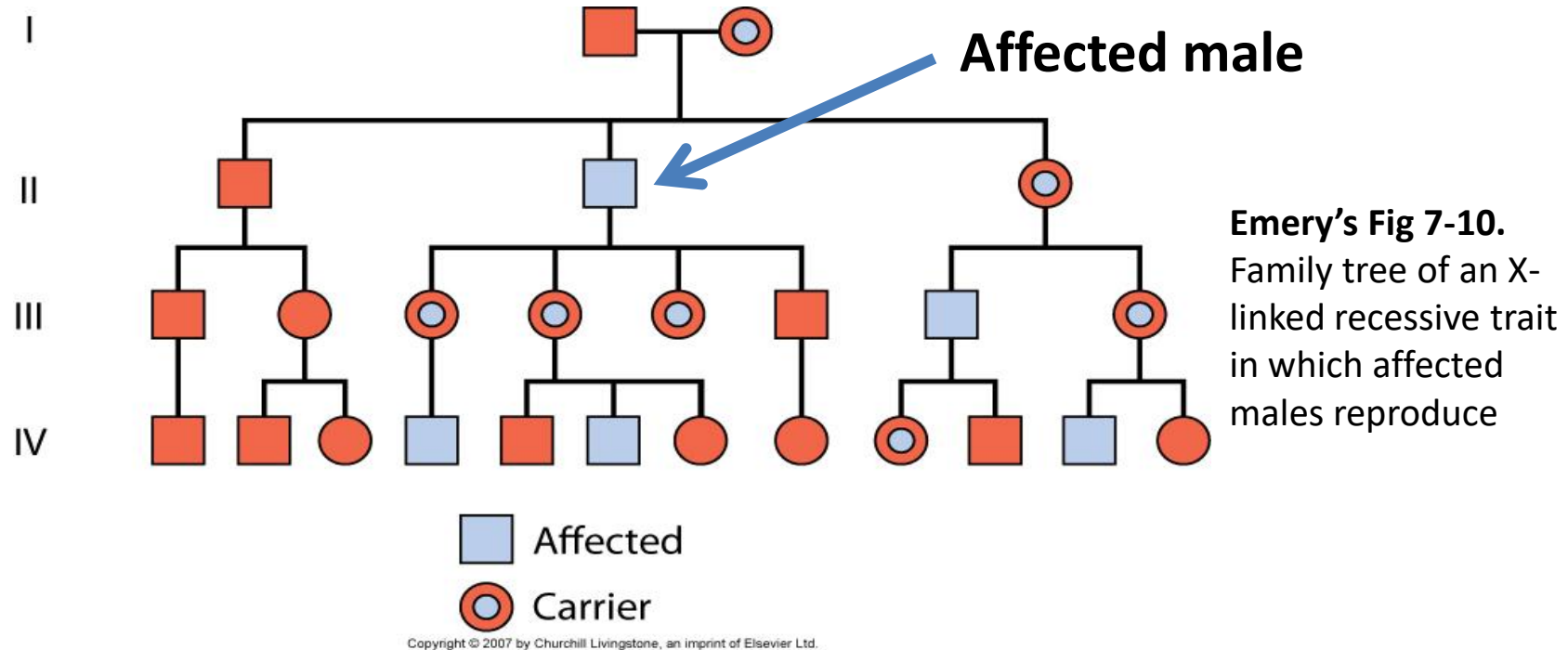
Hemophilia A – Intron inversion



Intron inversion (40% of mutations) disrupts FVIII gene

- (A) Factor VIII gene and repetitive non coding sequences (Int22h) and exons shown as filled boxes
- (B) Mis-alignment during Homologous recombination and part of factor VIII gene is lost
- (C) Completely disrupted factor VIII gene: **exons 1 to 22** are displaced towards the telomere and oriented opposite to their normal orientation. A contiguous factor VIII gene transcript from exon 1 to 26 is no longer possible and severe hemophilia A results

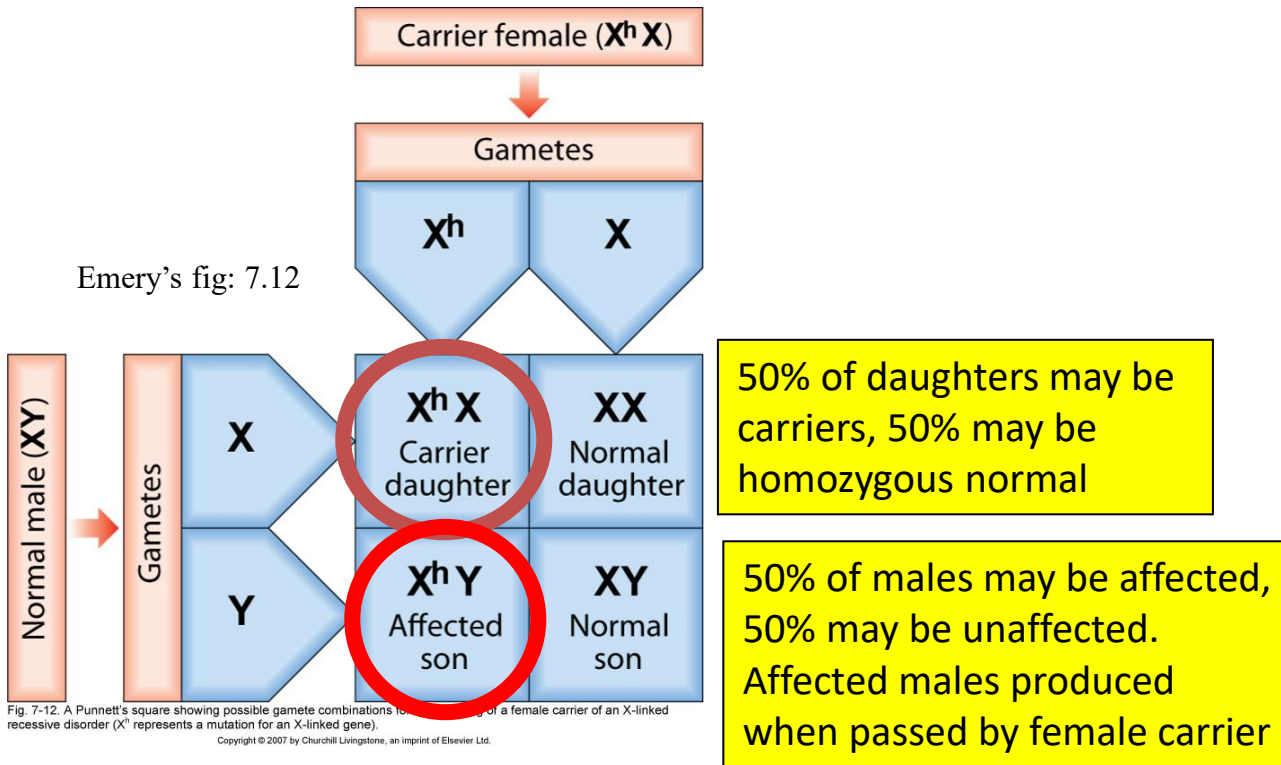
Inheritance Pattern for hemophilia



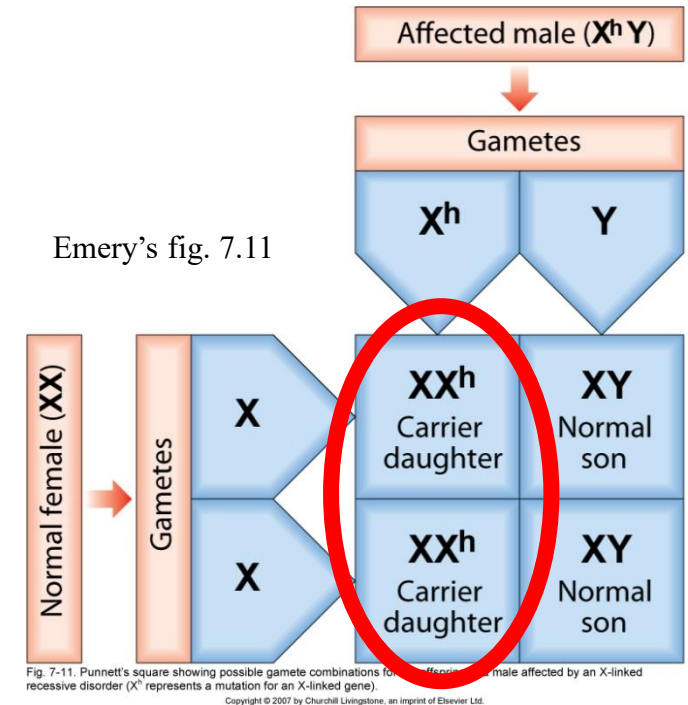
- Inheritance is Affected male → Carrier female → Affected male (Skipped generations)
- No male-to-male transmission. A son cannot inherit mutant allele from his father.
- Carrier mother has 50% chance of passing the faulty X-chromosome to her daughter
- Affected father always passes on affected gene to all his daughters (**Obligate carriers**)

Recurrence risk for a normal father (XY) and carrier mother (Xx*)

Males are more vulnerable as they have only one X chromosome (Males are Hemizygous)



Recurrence risk for affected father (x*Y) and normal mother (XX)



All daughters are **obligate carriers**. Only carrier females produced when passed by male. All sons are normal

Phenotype of carrier females

- Usually asymptomatic since **random X inactivation/ Lyonization** results in approximately equal proportions of somatic cells in which either normal X or mutant X chromosome is active, and clotting factor synthesis is adequate for hemostasis.
- Some carrier females have increased bleeding tendency
(Manifesting heterozygote)
 - Due to **skewed X-chromosome inactivation** leading to a higher percentage of inactivation of normal X-chromosome

Treatment

- **Recombinant factor VIII (or IX)** replacement therapy
 - Antibodies against recombinant protein in ~10% of patients
- Hemophiliacs received clotting factor from pooled blood
 - Risk of HIV and hepatitis B and C via contaminated blood (as donor blood was not rigorously screened)
 - More than 50% of hemophiliacs contracted HIV/ hepatitis from transfusion of infected blood (before testing for HIV was routinely practised)